



Hochschule
für Wirtschaft und Umwelt
Nürtingen-Geislingen

A Comparison Between VVIX and VIX

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Data Science in Finance: Financial Econometrics

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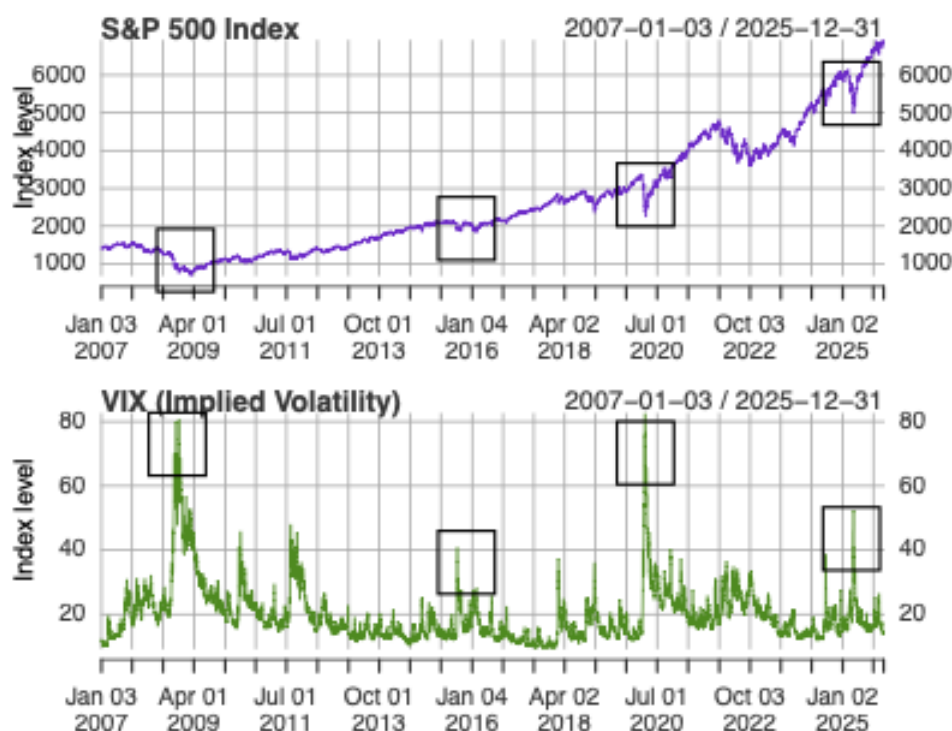
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1 Dataset and Objective

This analysis looks at the volatility dynamics of the S&P 500 by focusing on the VIX and the VVIX and comparing their behavior over the same period. The goal is to study how the VVIX responds to changes in the VIX and to see if its movements are more pronounced and persistent, especially during periods of market stress. The analysis is conducted on daily observations of the VVIX and VIX indices, both obtained from Yahoo Finance. The data are from 01-01-2007 to 31-12-2025, covering different market phases and several periods of pronounced financial stress. The analysis proceeds in three steps: first, price dynamics and log returns are examined. Second, key stylized facts are analyzed, including volatility clustering, autocorrelation patterns, leverage effects, non-normality, and VaR violations. Third, volatility is modeled using an EGARCH model, and the joint dependence between VIX and VVIX is studied using a Student-t copula, with particular attention to tail behavior. A specific focus is given to the COVID-19 period to evaluate volatility amplification effects.

2 Data and Price Dynamics

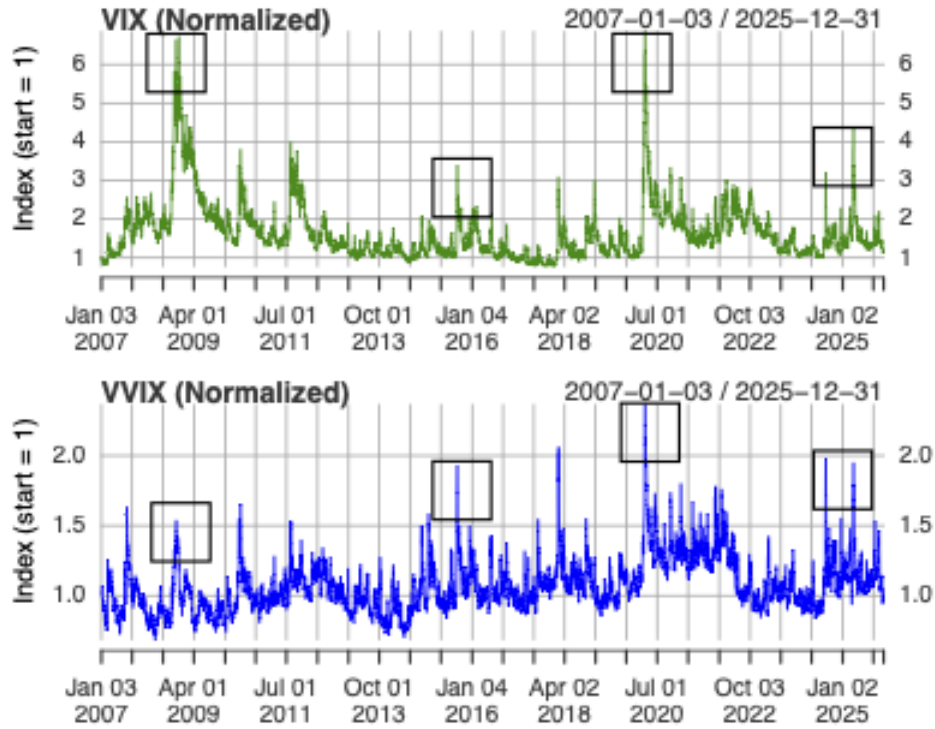
2.1 S&P 500 and VIX



The first figure shows the evolution of the S&P 500 index together with implied volatility, measured by the VIX, over the period 2007-2025. The two panels clearly display the inverse relationship between equity prices and market uncertainty. When the S&P 500 experiences declines, the VIX simultaneously records strong spikes. The highlighted periods correspond to major stress episodes. During the 2008-2009 financial crisis, the S&P 500 drops dramatically while the VIX rises to extreme levels, reflecting a sudden increase in uncertainty and fear. A similar pattern is visible during the Covid-19 shock in 2020,

where the rapid collapse in equity prices is matched by one of the highest volatility spikes in the entire sample. Additional stress phases, such as the 2015-2016 turbulence, also show moderate but clearly visible increases in the VIX. Outside these episodes, the S&P 500 follows a relatively stable upward trend, while the VIX remains at lower levels, typically fluctuating within a certain range.

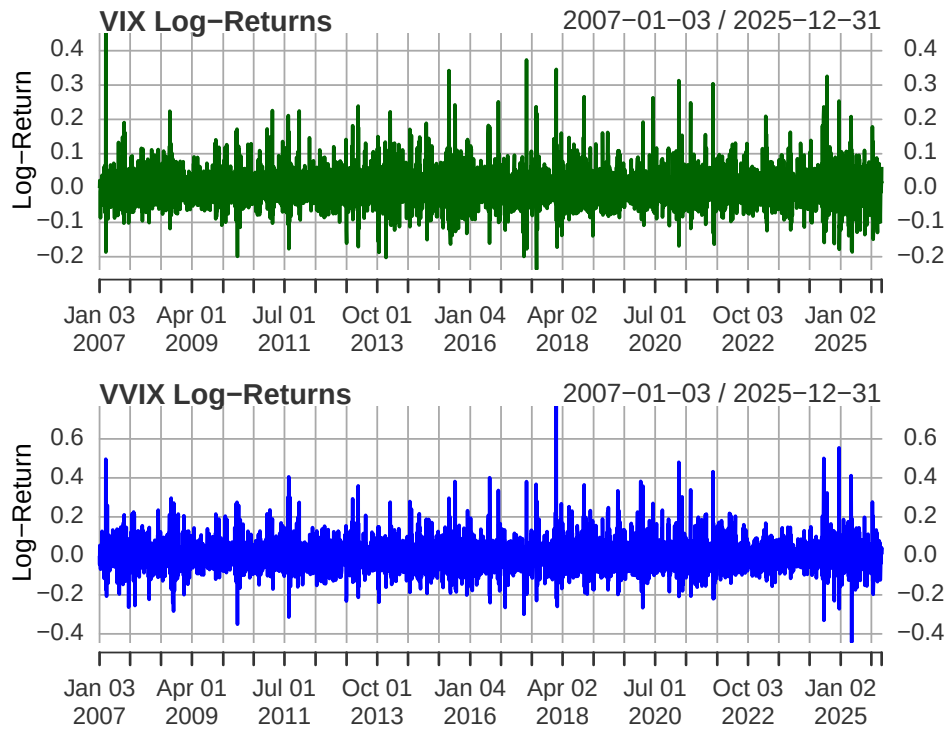
2.2 VIX and VVIX



In the second figure there is a comparison of the normalized dynamics of the VIX and the VVIX. During major crisis episodes, the VIX exhibits pronounced spikes, which are generally accompanied by increases in the VVIX. However, the magnitude and timing of VVIX movements differ from those of the VIX. In particular, the VVIX tends to react more strongly when implied volatility displays sudden and irregular movements, suggesting that it captures the instability of volatility itself rather than its level. Looking at the highlighted periods, it is clear that some spikes in the VIX are followed by even more persistent or amplified reactions in the VVIX. This indicates that when uncertainty about market volatility increases, the variability of volatility expectations rises disproportionately. Moreover, in calmer periods the VIX may return relatively quickly to its baseline, while the VVIX often remains elevated for longer, reflecting sustained uncertainty about future volatility dynamics.

3 Stylized Facts

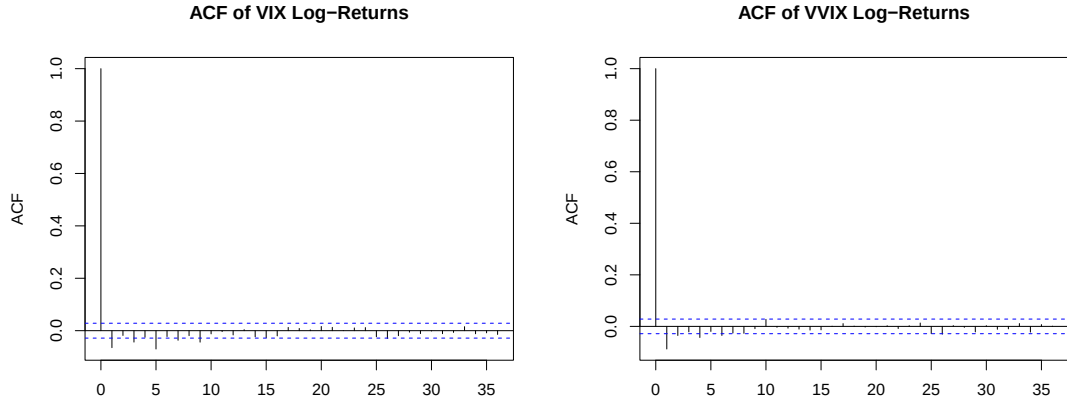
3.1 Volatility Clustering



Comparing the log returns of the VVIX and VIX indices, both series clearly exhibit volatility clustering, with periods of relatively calm dynamics alternating with phases of intense fluctuations. Large movements tend to be followed by further large movements, while tranquil periods persist for some time, confirming the presence of persistent second-moment dynamics in both indices. However, the VVIX log-returns display substantially larger amplitudes and more frequent extreme observations compared to the VIX. The spikes in VVIX are sharper and often reach significantly higher positive and negative values within short time intervals. In contrast, VIX log-returns, although volatile, appear more contained and less explosive. This difference suggests that VVIX reacts more aggressively to market shocks than implied volatility itself. The VVIX therefore shows higher variability, stronger clustering of extreme values, and a more unstable pattern overall, especially during stress episodes.

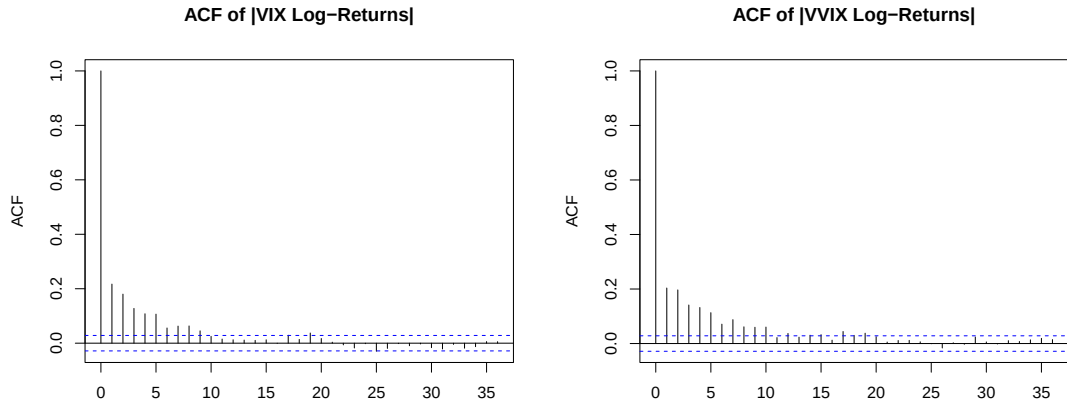
3.2 Autocorrelation Analysis

3.2.1 ACF of Log Returns



The autocorrelation functions of VIX and VVIX log-returns show that autocorrelations at all lags are small and remain within the confidence bands. This indicates that daily changes in both VIX and VVIX exhibit limited linear dependence on their past values.

3.2.2 ACF of |Log Returns|



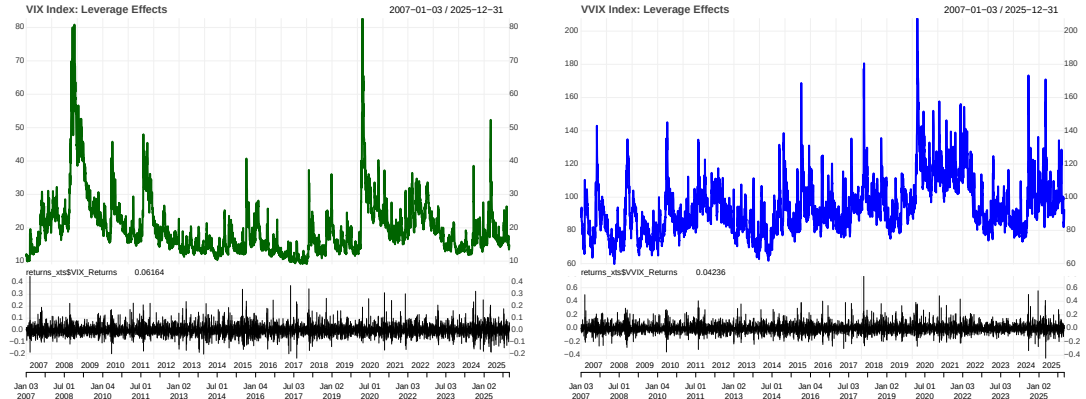
A different pattern emerges when considering absolute log-returns. Both VIX and VVIX display positive and persistent autocorrelation at short lags, indicating volatility clustering. However, the effect is more pronounced for the VVIX, whose autocorrelations are larger and decay more slowly.

3.2.3 Autocorrelation Analysis Results

Comparing the two sets of results, a clear difference emerges between linear dependence and volatility persistence. For both VIX and VVIX, the ACF of log-returns shows no significant autocorrelation, confirming that daily changes are largely unpredictable in their direction. When looking at absolute log-returns, both series exhibit strong and positive autocorrelation at short lags, indicating persistent volatility clustering. This effect is clearly stronger for the VVIX: its autocorrelations are higher and decay more slowly.

3.3 Leverage Effect

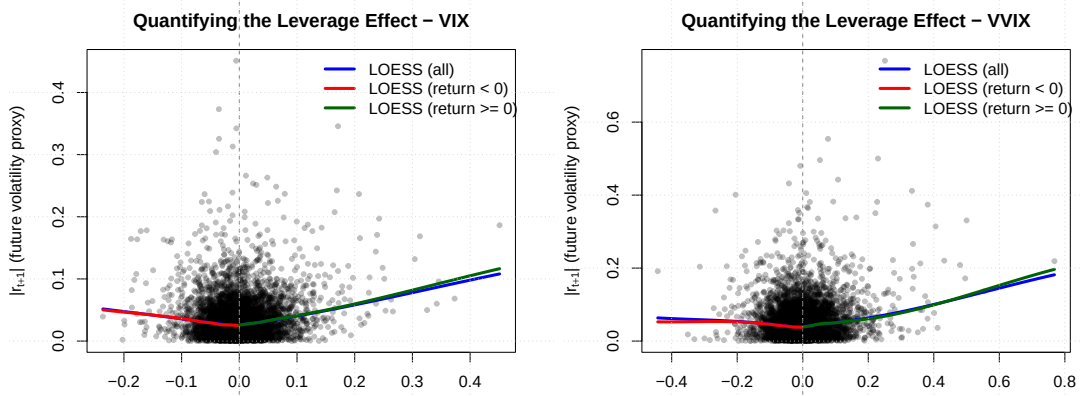
3.3.1 Evidence of Leverage Effect



The leverage effect appears in both indices but with different characteristics. In the VIX, the asymmetry between returns and volatility is mainly visible during periods of market stress, while in the VVIX the negative relationship appears more pronounced and more frequent even outside of crisis periods. At a descriptive level, the VVIX exhibits larger volatility reactions following negative returns compared to the VIX.

3.3.2 Quantifying Leverage Effect

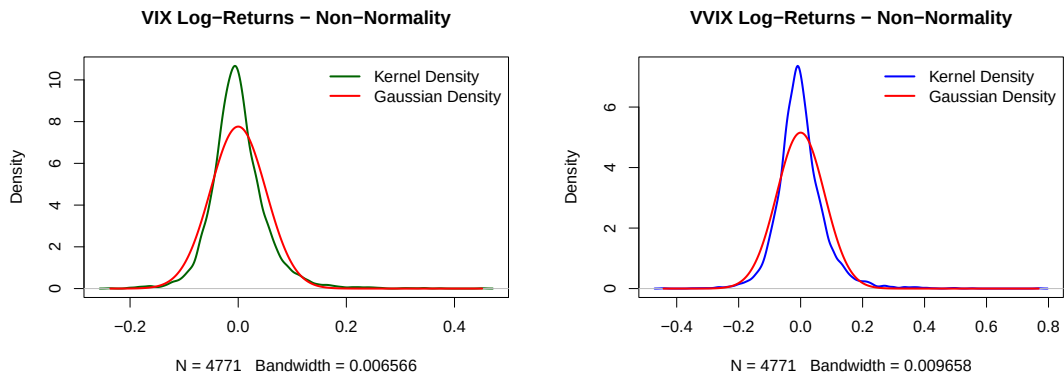
These graphs are included to directly assess whether negative returns are followed by larger volatility reactions than positive returns. For this purpose, current returns are plotted against the absolute value of next-period returns, used as a simple proxy for future volatility. By separating negative and positive returns and smoothing the relationship using LOESS curves, the graphs allow a direct visual comparison of volatility responses for both VIX and VVIX.



Each graph shows the relationship between current returns r_t and future volatility, measured by the absolute value of next-period returns $|r_{t+1}|$. The LOESS lines show how future volatility reacts differently to positive and negative returns. A clear asymmetry appears in both cases: negative returns are generally followed by higher future volatility than positive returns of similar size. For the VIX, this difference is relatively small and mainly concentrated around returns close to zero. In contrast, for the VVIX, the gap

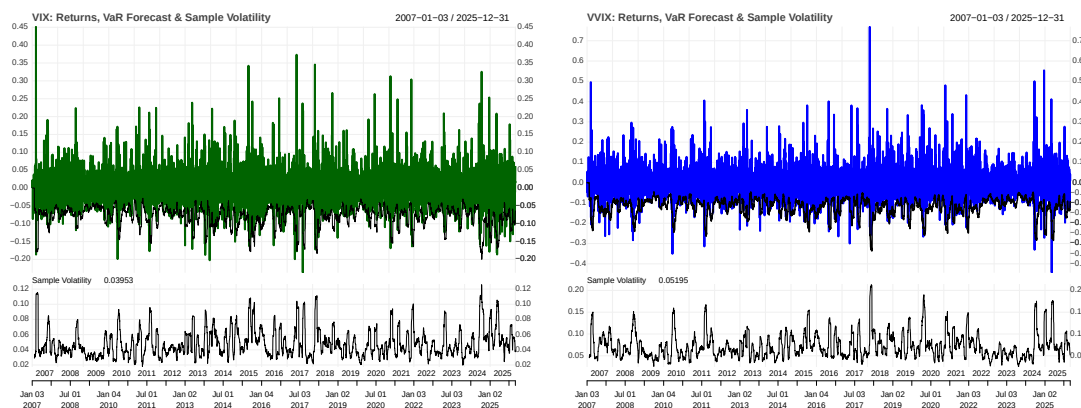
between negative and positive returns is larger and remains visible over an expansive range. The VVIX reacts more strongly to negative shocks than the VIX.

3.4 Non Normality



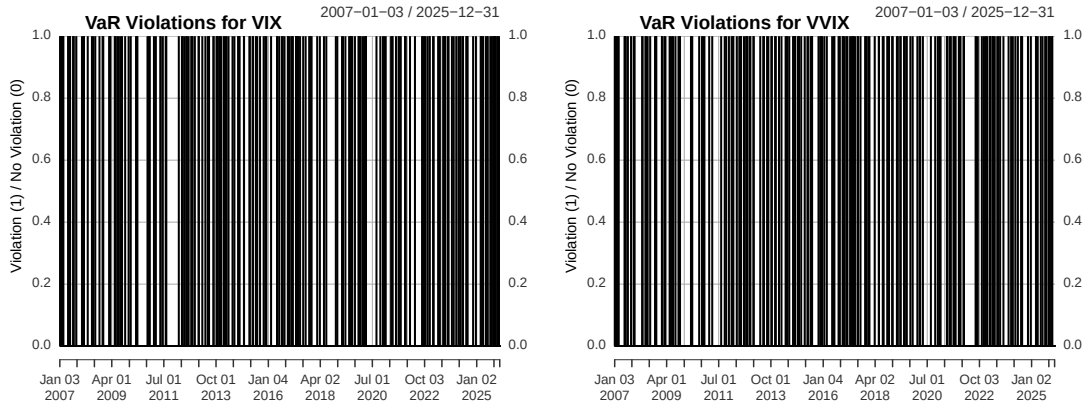
The comparison of the empirical density of VIX and VVIX log-returns is estimated using kernel density, with a Gaussian distribution having the same mean and variance. In both cases, the empirical distribution deviates from normality, particularly in the tails. Non-normality is more pronounced for the VVIX, which exhibits a higher frequency of extreme returns compared to the VIX.

3.5 Forecasting Volatility with Sample Volatility



The comparison between the VIX and the VVIX highlights clear differences in terms of return amplitude, volatility, and risk. The VVIX exhibits more extreme and frequent fluctuations, resulting in a higher sample volatility, while the VIX shows more moderate movements. During periods of market stress, the VaR estimated for the VVIX expands substantially, indicating a strong amplification of risk, while the VIX responds in a more stable and orderly manner. The results suggest that the volatility of volatility index (VVIX) reacts more intensely to market shocks than the level of volatility (VIX) itself.

3.6 VaR Violations



The figures show the days in which the **5% VaR** is violated, meaning days when observed losses are larger than the estimated risk level. For the VIX, violations mostly appear during periods of strong market stress conditions. For VVIX, violations occur more frequently and tend to cluster more closely over time, even outside major crisis periods. This shows that the risk linked to the VVIX is less stable and more difficult to capture than the risk measured by the VIX. For the VVIX, violations occur more frequently and are distributed more unevenly across the sample. They tend to cluster not only during major crises but also during intermediate phases of heightened uncertainty. This suggests that the VVIX is more difficult to model accurately and that extreme movements are less confined to clearly defined stress episodes.

4 Volatility Modeling

In this chapter, some of the most important GARCH(1,1) models are estimated on VIX and VVIX returns. The models considered are GARCH, GJR-GARCH and EGARCH, all under Student-t innovations. The comparison allows to assess how different specifications describe the volatility dynamics of the two series.

Table 1: VIX returns			Table 2: VVIX returns		
Model	AIC	BIC	Model	AIC	BIC
EGARCH(1,1)	-3.342	-3.334	EGARCH(1,1)	-2.566	-2.558
GJR-GARCH(1,1)	-3.337	-3.329	GJR-GARCH(1,1)	-2.555	-2.547
sGARCH(1,1)	-3.325	-3.318	sGARCH(1,1)	-2.532	-2.525

Compared to the other models, **EGARCH(1,1)** achieves the lowest AIC and BIC in both cases, indicating a superior ability to capture the volatility dynamics of the two series. The EGARCH model is selected for the consequent analysis.

4.1 EGARCH

The EGARCH(1,1) formula is:

$$\log(h_t) = \omega + \beta \log(h_{t-1}) + \alpha \left(\frac{|\varepsilon_{t-1}|}{\sqrt{h_{t-1}}} - \mathbb{E} \left[\frac{|\varepsilon_{t-1}|}{\sqrt{h_{t-1}}} \right] \right) + \gamma \frac{\varepsilon_{t-1}}{\sqrt{h_{t-1}}} \quad (1)$$

Parameter	VIX	VVIX
μ	-0.00117	-0.00144
ω	-0.59716	-0.31760
α	0.18882	0.23689
β	0.90357	0.94295
γ	0.15424	0.07283
shape	4.61	4.91

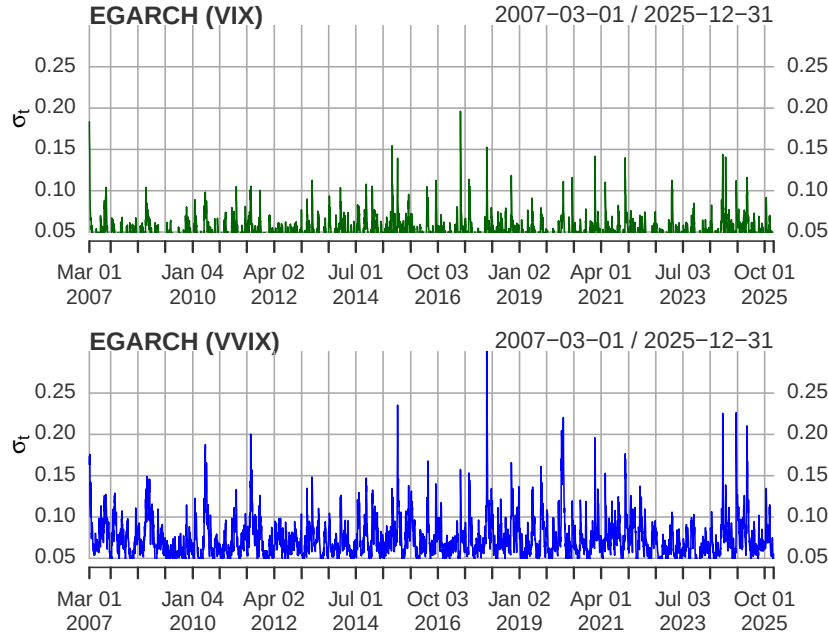
4.1.1 EGARCH: Results

- **Mean dynamics (μ).** The estimated mean returns are close to zero for both indices. This show that the relevant dynamics are motivated by time-varying volatility rather than by predictable movements in the mean.
- **Baseline volatility (ω).** The intercept is negative for both series. Its size differs between the two indices, showing that the VIX and the VVIX have different baseline levels of volatility.
- **Shock magnitude (α).** The shock parameter is larger for the VVIX than for the VIX, meaning that VVIX reacts more strongly to unexpected shocks. Even small changes in market conditions can lead to large shifts in uncertainty about future volatility.

- **Volatility persistence (β).** Both indices show very high persistence, with β values close to one. However, persistence is clearly stronger for the VVIX, meaning that shocks to volatility of volatility fade away more slowly.
- **Asymmetry (γ).** The asymmetry parameter is positive and statistically significant for both series, confirming that negative shocks generate stronger volatility responses than positive shocks of similar magnitude. For the VVIX, this asymmetric response contributes to prolonged periods of elevated volatility following adverse market events.
- **Distributional shape (shape).** The estimated shape parameters show that returns for both indices have heavy tails. This means that extreme movements happen more often than what a normal distribution would suggest, especially for the VVIX.

The results show clear differences between the VIX and the VVIX. The VVIX reacts more strongly to shocks, shows higher persistence, and displays a more pronounced asymmetry than the VIX. This means that uncertainty about volatility tends to last longer and increases especially after negative shocks. These features are also clearly visible in the following figure.

4.1.2 EGARCH:Graph

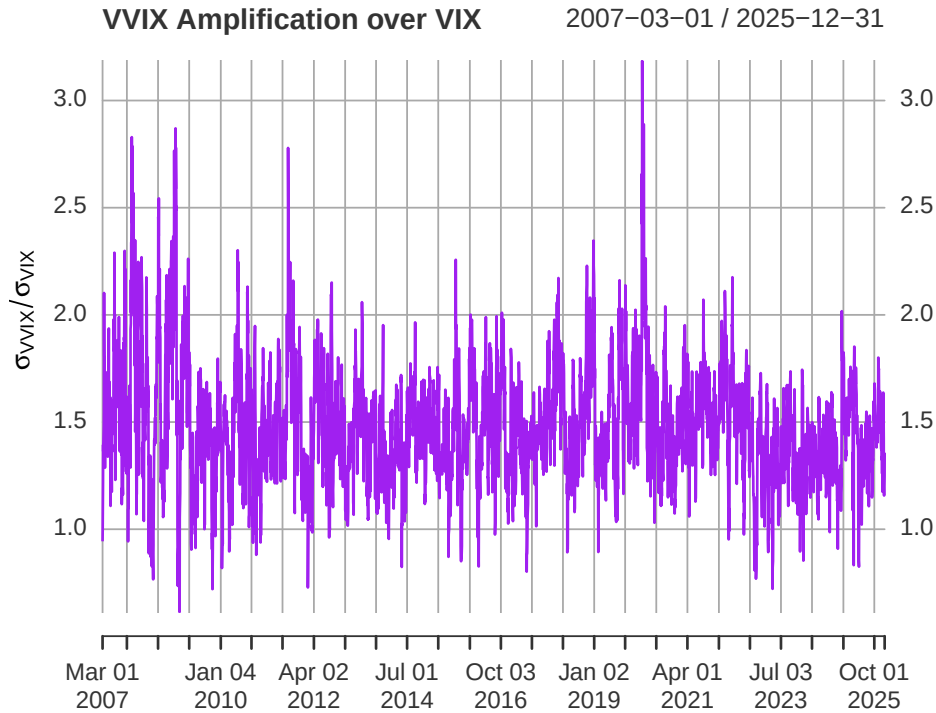


The two indices exhibit volatility spikes concentrated in the same periods, indicating strong co-movement during market stress phases. Peaks are clearly aligned in crisis episodes, confirming that shocks affecting market volatility are immediately reflected in volatility-of-volatility dynamics. However, the conditional volatility estimated for the VVIX is systematically higher than that of the VIX. The VVIX series shows larger spikes, higher dispersion, and more persistent elevated levels after major stress events. This suggests that uncertainty about future volatility reacts more aggressively and remains elevated for longer compared to the volatility of equity returns itself. The EGARCH

dynamics also reveal asymmetric behavior during stress periods. In sharp market deteriorations, volatility increases rapidly and the amplitude of the spikes is significantly larger than during calm phases. After extreme events, volatility does not immediately revert to pre-crisis levels but declines gradually, indicating persistence in the conditional variance. This pattern is more pronounced for the VVIX, where clusters of high volatility are longer and more intense.

4.1.3 VVIX amplification over VIX

To better understand how volatility expectations react relative to market volatility itself, the amplification ratio is introduced. It is defined as the conditional volatility of the VVIX divided by the VIX level. The VIX measures the level of implied volatility in the equity market, while the VVIX reflects the volatility of that volatility. Looking at the two series separately does not fully capture how strongly volatility-of-volatility reacts compared to the underlying volatility. The amplification ratio provides a direct measure of this relative sensitivity. When the ratio is above one, changes in volatility expectations are proportionally stronger than changes in the VIX itself. This allows to identify periods in which uncertainty about future volatility increases more aggressively than market volatility, especially during stress episodes. For this reason, the ratio is a useful extension of the previous analysis and helps to quantify how market stress propagates from volatility to volatility-of-volatility dynamics.



The "VVIX Amplification over VIX" figure shows the ratio between the conditional volatility of the VVIX and the VIX over time. For most of the sample, this ratio stays above one, meaning that the volatility of volatility is usually higher than the volatility of the VIX itself. This indicates that uncertainty about future volatility tends to be stronger than uncertainty about market returns. During periods of market stress, the ratio increases sharply, showing that the VVIX reacts much more strongly than the VIX

when volatility rises suddenly. These spikes suggest an amplification effect: when market conditions deteriorate, not only does volatility increase, but uncertainty about volatility grows even faster. Over calmer periods, the ratio remains more stable but still above one, confirming that the VVIX consistently reflects a higher level of instability than the VIX.

4.1.4 Ljung Box Test

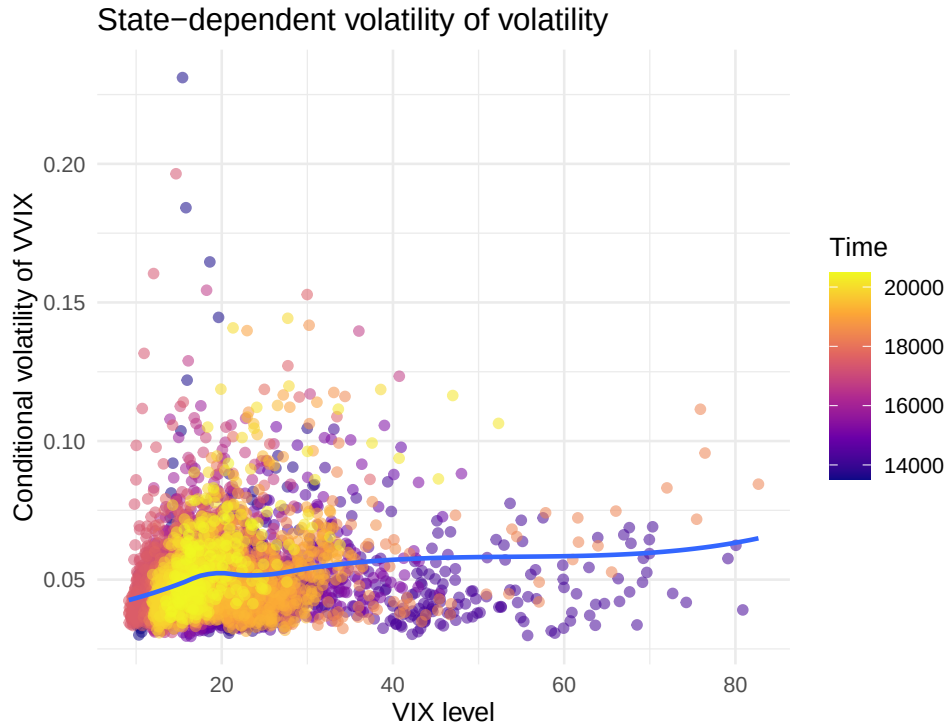
Series	Lag 10 p-value	Lag 20 p-value
VIX	0.9589	0.8426
VVIX	0.8238	0.3748

The Ljung–Box tests on squared standardized residuals do not provide evidence of remaining serial dependence for either VIX or VVIX at lag 10 or lag 20. All reported p-values are well above conventional significance thresholds, suggesting that no relevant autocorrelation structure remains in the residuals after the EGARCH specification.

- For the VIX, the p-values of 0.9589 (lag 10) and 0.8426 (lag 20) are particularly high, indicating that the model effectively captures the volatility clustering present in the data. There is no indication of unmodeled persistence in the conditional variance.
- For the VVIX, the p-values are slightly lower but still clearly non-significant (0.8238 at lag 10 and 0.3748 at lag 20). Even at longer horizons, the null hypothesis of no serial correlation cannot be rejected. This confirms that the EGARCH dynamics are sufficiently flexible to account for the stronger and more frequent volatility spikes observed in the VVIX.

The diagnostic results support the adequacy of the estimated models. The absence of residual autocorrelation in squared standardized residuals indicates that the main volatility dynamics have been successfully captured.

5 State-dependent volatility of volatility



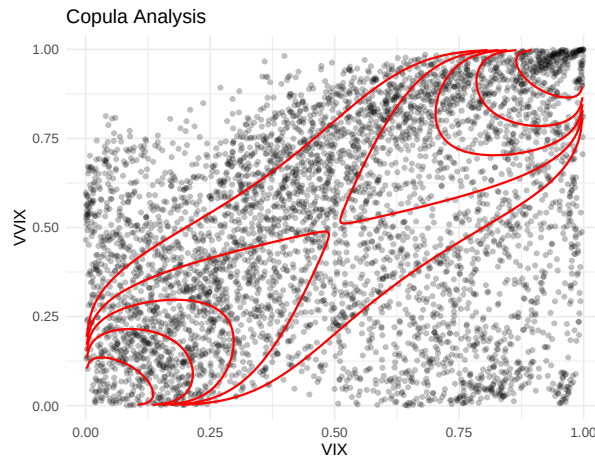
This plot links the VIX level to the conditional volatility of the VVIX estimated with the EGARCH model. Each point represents one trading day. The x-axis reports the level of the VIX, while the y-axis shows the estimated conditional volatility of the VVIX for the same day. The color scale captures the time dimension, allowing to visually separate different market phases.

- The first clear result is that the relationship is positive. When the VIX is low (around 10–20), the conditional volatility of the VVIX is also relatively low and concentrated in a narrow band. The dispersion of points in this region is limited, suggesting that VVIX remains stable during calm market conditions.
- As the VIX increases (above 30–40), the dispersion of the VVIX becomes much wider. The graph shows more extreme values and stronger variability. This means that not only is market volatility higher, but the uncertainty about future volatility becomes more unstable. The smooth regression line confirms this upward trend: higher VIX levels are associated with higher VVIX levels
- In very high VIX regimes (above 60), the conditional volatility of the VVIX increases further and becomes much less predictable. These observations are consistent with crisis periods, where shocks propagate more strongly into expectations about volatility.
- Another important observation is the heteroskedastic pattern in the scatter. At low VIX levels, the cloud of points is tight and compact. At high VIX levels, the cloud becomes more dispersed vertically. This suggests that extreme stress periods are characterized by instability in volatility dynamics.

6 Copula Analysis

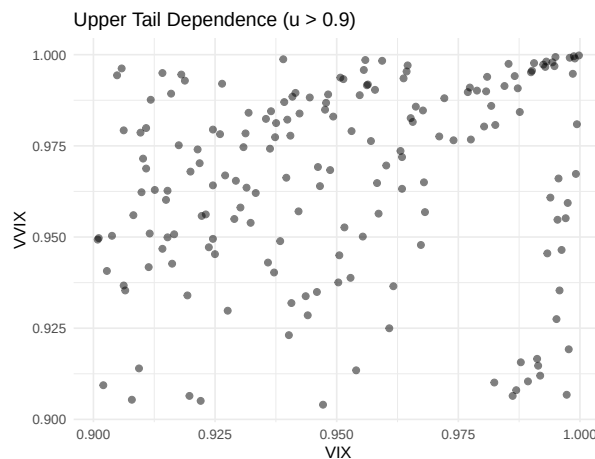
A Student-t copula is estimated to analyze the dependence between VIX and VVIX. Then an analysis on the upper tail is made to see how VVIX reacts compared to VIX during the market-stress scenario.

6.1 Student-t Copula



The figure shows the Student-t copula between VIX and VVIX, allowing for symmetric tail dependence. Density contours indicate stronger co-movements when both indices take extreme values. This suggests that VVIX depends on VIX not only on average, but especially during extreme market conditions.

6.1.1 Upper Tail Analysis

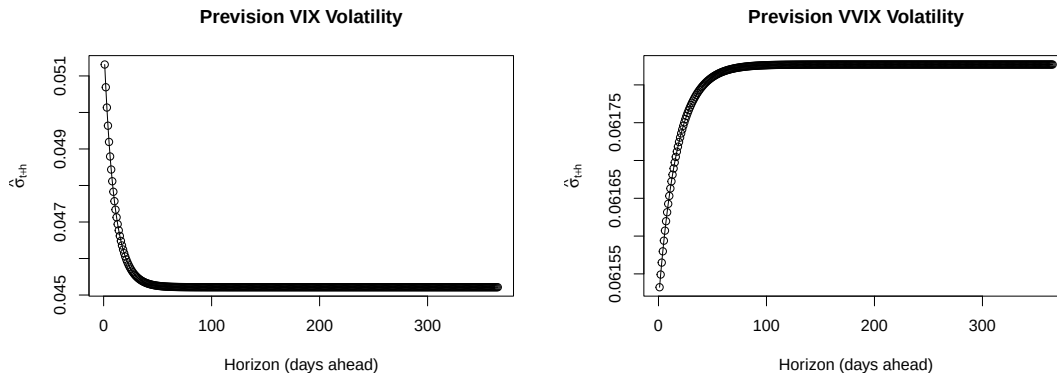


The figure focuses on the upper tail of the joint distribution, highlighting extreme market stress episodes. When VIX reaches very high levels, VVIX is frequently elevated as well, indicating joint extreme realizations. This pattern shows that volatility-of-volatility increases systematically with market fear. VVIX appears to be strongly dependent on VIX during stress conditions.

7 Forecasting Volatility

This section computes 365-day ahead conditional volatility forecasts for both VIX and VVIX using the estimated the EGARCH model. The forecasts are generated starting from the last available observation in the sample.

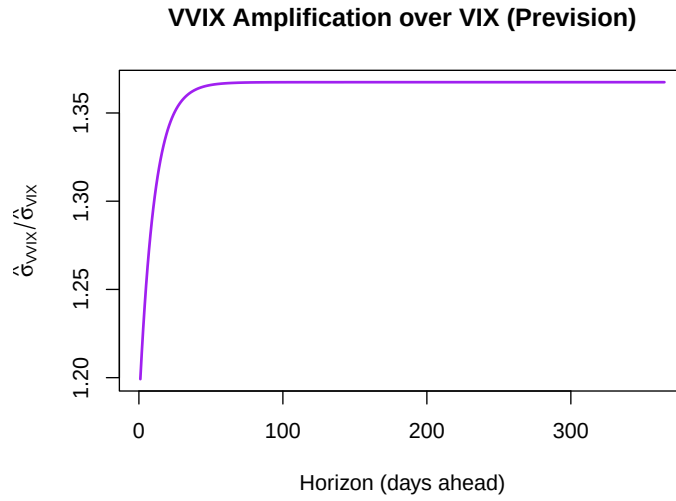
7.1 Volatility Prevision



The EGARCH volatility forecasts show a clear adjustment pattern for both indices. For the VIX, the forecast starts from a relatively elevated volatility level and declines quickly during the first forecast horizons. After approximately 40 to 50 days, the curve becomes almost flat and remains stable for the rest of the horizon. This indicates that the current volatility level is above the level implied by the model's dynamics and is expected to gradually decrease over time. For the VVIX, the adjustment pattern differs. The conditional volatility starts from a lower level and gradually increases over the forecast horizon before stabilizing. The convergence is slower compared to the VIX, and the forecasted volatility remains consistently higher across the horizon. This reflects stronger persistence in the VVIX dynamics and a more prolonged adjustment process relative to implied volatility.

- The key difference between the two forecasts is in the speed of convergence and in the level at which volatility stabilizes. The VVIX converges more slowly and remains at a higher forecasted volatility level across the horizon. This indicates that VVIX adjusts more gradually and stays elevated for longer compared to VIX.

7.2 Amplification Ratio Prevision

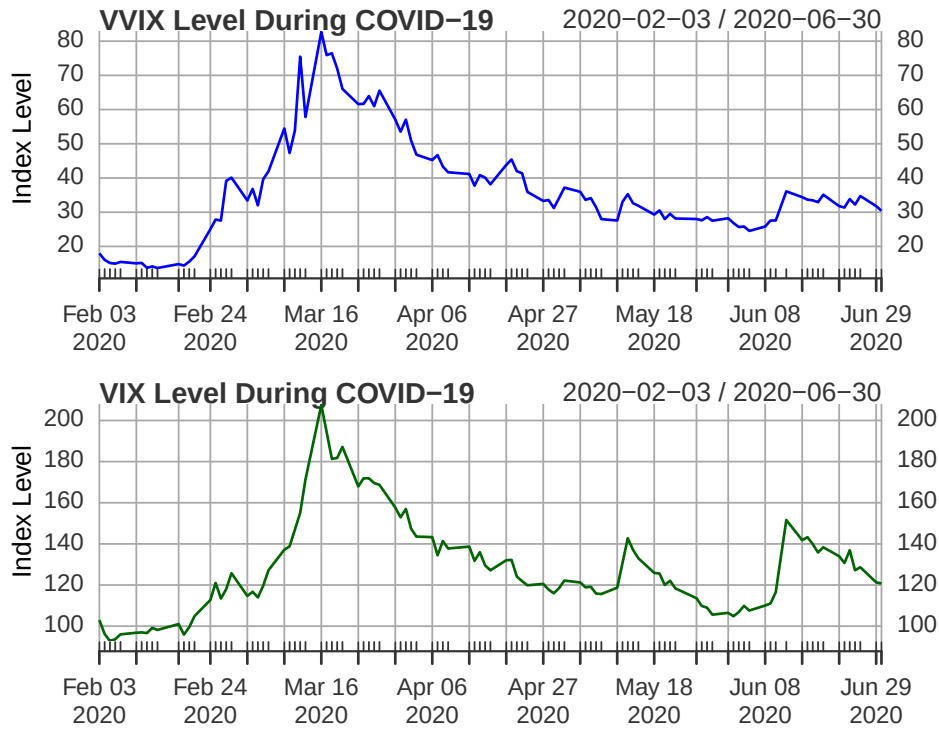


The forecasted amplification ratio shows a clear and stable pattern across the entire forecast horizon. The ratio starts at a level slightly above 1.20 and increases rapidly during the first forecast periods, before stabilizing around 1.35. This initial increase reflects the different adjustment dynamics of the two indices: while VIX volatility declines toward its equilibrium level, VVIX volatility increases toward a higher long-run level. Most importantly, the ratio remains strictly above one for all forecast horizons. This indicates that VVIX is structurally higher than implied volatility, not only during stress periods but also in the long-run equilibrium implied by the model.

8 Spike Analysis: COVID Period

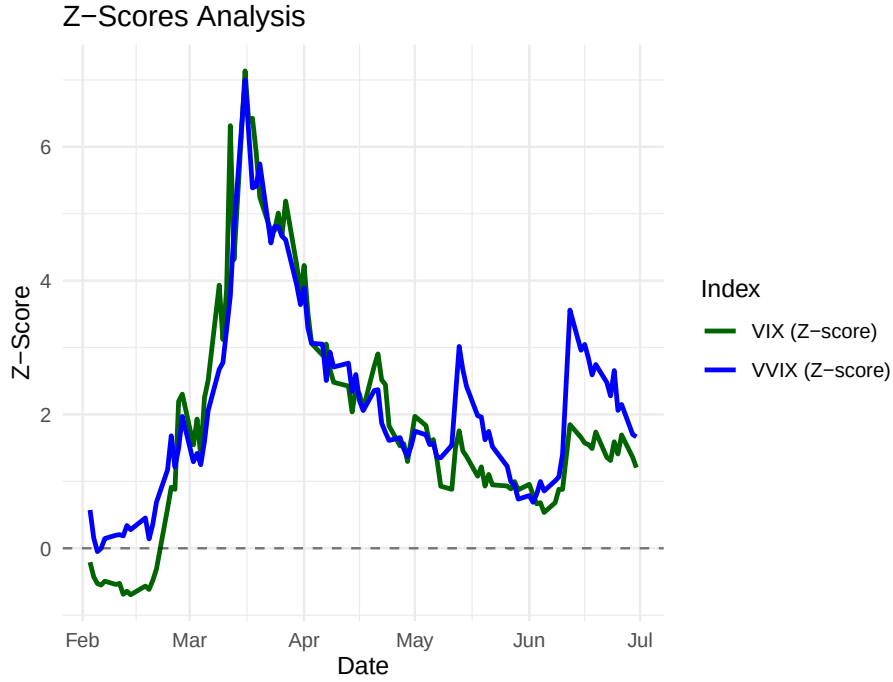
This section focuses on the COVID-19 period as an extreme stress episode. The objective is to analyze whether the VVIX amplifies the shock observed in the VIX during the COVID-19 crisis, exhibiting more intense and more persistent spikes. The analysis compares the behavior of the two indices around the event.

8.1 Price Levels during COVID



These graphs show the evolution of the VVIX and the VIX around the COVID-19 period. Before the arrival of COVID-19, both indices remain at relatively stable levels. Between late February and mid-March 2020, both series react to the shock: the VIX increases rapidly, while the VVIX rises even more strongly in relative terms. Around the peak, the VVIX shows larger and more irregular movements. After mid-March, the VIX starts to decline, while the VVIX remains elevated and volatile for a longer period, showing repeated oscillations.

8.2 Z-Scores Analysis

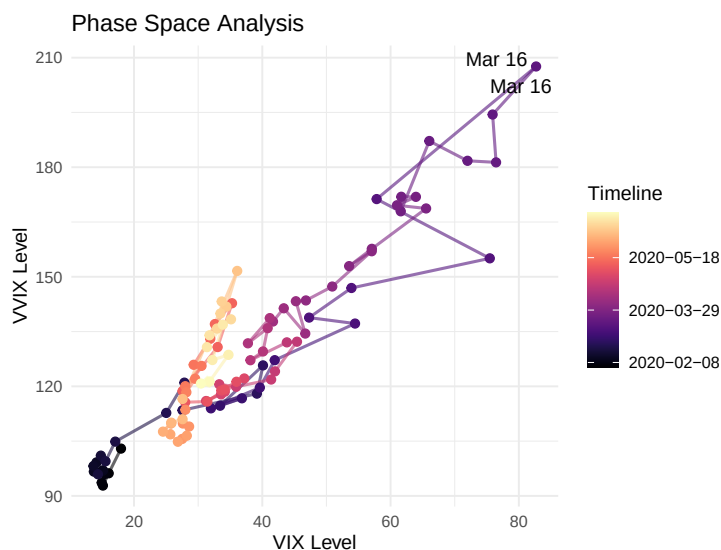


This analysis compares the standardized spike intensity of the VIX and the VVIX using Z-scores. The Z-score measures how many standard deviations an observation deviates from its historical mean, providing a standardized measure of relative intensity.

$$z_t = \frac{x_t - \mu}{\sigma} \quad (2)$$

This transformation allows a direct comparison of how extreme each movement is relative to its own historical distribution. During the COVID-19 crisis, VVIX exhibits substantially higher Z-scores than the VIX, indicating more intense deviations from its mean. In the peak phase of the shock, VVIX reaches levels exceeding those of the VIX by several standard deviations. While both indices spike strongly in March, the magnitude of the VVIX spike is clearly larger, showing that VVIX reacts more aggressively than market volatility itself. After the initial peak, the VIX Z-score declines quickly, while the VVIX remains elevated for a longer period.

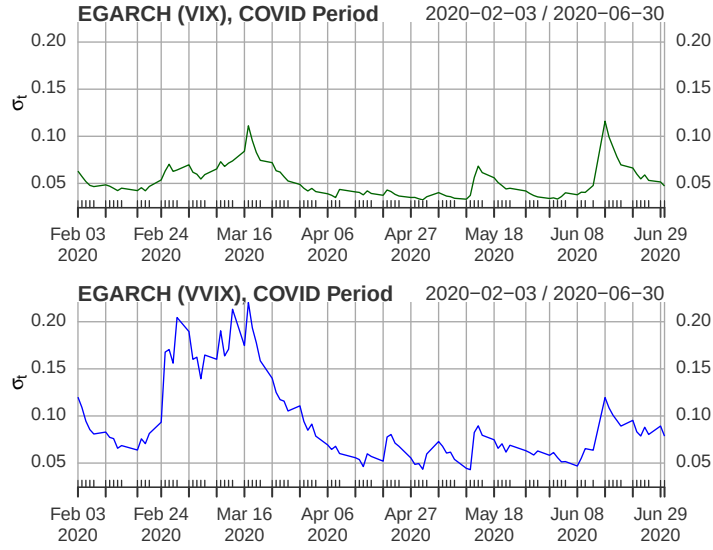
8.3 Phase Space Analysis



The phase space plot shows how VIX and VVIX moved during the COVID shock. At the beginning, the path goes almost straight up. VVIX jumps quickly while VIX is still increasing. Around mid-March, VVIX reaches extreme levels even before VIX hits its maximum. Even when VIX is close to its peak and starts stabilizing, VVIX stays very high and keeps spiking. This suggests that even if market fear was already extreme, uncertainty about future volatility was still increasing. During the recovery, VIX comes down faster, while VVIX declines more slowly. This creates a kind of loop in the graph. The upward move is sharper than the downward one. VVIX amplifies the shock. It moves earlier, moves more, and stays elevated longer than VIX.

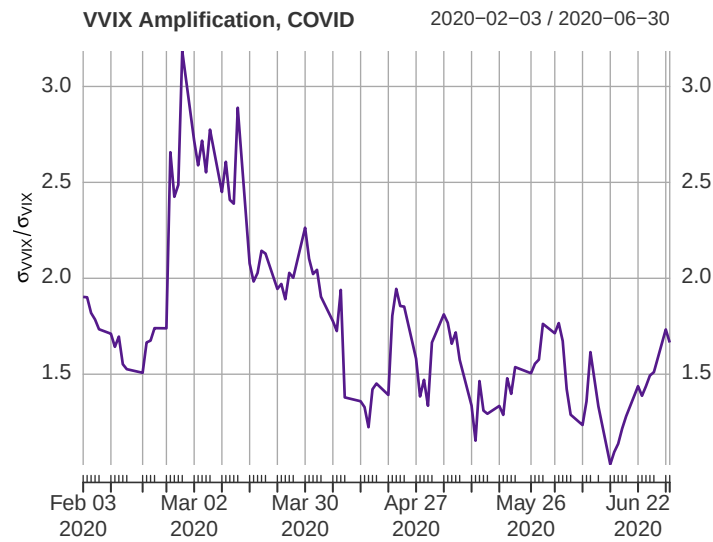
8.4 Volatility Analysis: COVID Period

8.4.1 E-Garch Graph, COVID Period



The figure shows the EGARCH-estimated conditional volatility for VIX and VVIX during the most intense phase of the COVID-19 shock (from February to June 2020). Both series exhibit a strong increase in mid-March, but the conditional volatility of the VVIX reaches levels nearly twice as high as that of the VIX, highlighting a clear amplification effect. Furthermore, while the VIX volatility declines relatively quickly after the peak, the VVIX remains elevated for a longer period and displays more pronounced fluctuations. This suggests that during the crisis not only did market volatility rise, but uncertainty about future volatility increased even more strongly and persistently.

8.4.2 VVIX amplification over VIX, Covid Period



The amplification ratio between the conditional volatility of the VVIX and that of the VIX during the COVID-19 shock (from February to June 2020) shows how VVIX am-

plifies the VIX shocks. The ratio remains consistently above one throughout the period, confirming that volatility-of-volatility is systematically higher than first-order implied volatility. Around mid-March, the ratio increases sharply, exceeding 2.5 and approaching 3, indicating a strong amplification mechanism at the peak of market stress. Although the ratio declines after the initial spike, it remains elevated relative to pre-crisis levels, suggesting that uncertainty about volatility persisted even as overall market volatility gradually stabilized.

9 Final Comparison

The comparison between VIX and VVIX reveals clear and systematic structural differences. While both indices respond strongly during market stress episodes, VVIX consistently exhibits larger return amplitudes, stronger volatility clustering, more pronounced asymmetry, and heavier tails.

- The EGARCH results confirm that VVIX reacts more aggressively to unexpected shocks and displays higher persistence. The higher shock parameter and stronger persistence imply that volatility-of-volatility not only increases faster, but also remains elevated for longer periods. This structural feature becomes particularly visible in crisis regimes, where the amplification ratio between VVIX and VIX rises sharply.
- The volatility forecasts reinforce this evidence. Both indices exhibit mean reversion toward a long-run equilibrium level, but the adjustment dynamics differ substantially. The VIX forecast converges relatively quickly, while the VVIX converges more slowly and to a structurally higher volatility level. The forecasted amplification ratio remains consistently above one and stabilizes at a level significantly above unity, indicating that volatility-of-volatility is not only episodically higher but structurally amplified in the long run.
- The COVID-19 episode provides a clear illustration of this mechanism. During the peak of market stress, VVIX volatility reached nearly twice the level of VIX volatility, and the amplification ratio exceeded 2.5. Even after VIX began to stabilize, VVIX remained elevated, indicating that uncertainty about future volatility persisted beyond the initial shock.
- The empirical evidence therefore shows that VVIX represents a distinct layer of risk. When market uncertainty increases, second-order volatility intensifies the reaction and prolongs instability, reinforcing stress dynamics across financial markets.